

Environmental Science and Sustainable Engineering

Individual Technical Report On High Carbon Emissions

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Abstract

The report discusses the problem of high carbon footprint which is one of the most pressing issues in terms of ecology and provides a solution to it through the development of a novel IT-based solution for calculation of GHG emissions – a localized multi-vector analytical engine (Energy, Mobility, Diet, Secondary Shopping, Waste Diversion), real-time EPA tree-offset equivalency tracking, and ISO 14001 goal-setting Carbon Footprint Calculator web portal.

Introduction

Emissions of carbon dioxide have reached record highs, contributing to global warming and ecological degradation. Although industry is largely responsible for these emissions, personal actions, such as the commute to work, and even diet, add significantly to the problem. Solutions to these problems involve local knowledge and resources that allow individuals to understand the relevant scientific data.

Problem Statement

Higher carbon emissions are contributing to climate anomalies. One of the reasons for which reduction in these emissions from consumers cannot be brought about is that there is no visibility, and people do not have any idea of the measurable effect their actions have on the environment.

Sustainability Analysis

- **Relevant SDGs:** This solution directly addresses **SDG 13 (Climate Action)** by improving education and awareness on climate mitigation, and **SDG 12 (Responsible Consumption and Production)** by encouraging users to optimize their resource use.
- **ISO 14001 Alignment:** Although it is usually used in business organizations, the basic concept of ISO 14001, which involves continuous environmental improvement via the "Plan-Do-Check-Act" process, is integrated into the suggested solution. In the suggested solution, users will plan, do, check, and act.
- **Life Cycle Assessment (LCA) Context:** The calculator will feature modules that determine the life cycle carbon emissions of consumer goods, enabling users to select products that have low cradle-to-grave carbon footprints.

Proposed Engineering/IT Solution

To solve high carbon emissions, the recommended solution is a dynamic Carbon Footprint Calculator. The solution will be an innovative responsive website developed using a current technology stack, such as React.js for the front end and Node.js for the back end.

- **Unique Features:**
 1. Setting reduction targets using the **Interactive Reduction Goal Simulator Slider** (0% to 50%).

2. Logging daily and monthly activity metrics across 5 distinct tabs.
3. Visualizing real-time analytics through the **Dual Charting Engine** (Category Doughnut Chart vs. Paris Agreement Benchmark Bar Chart).
4. Executing localized, sector-specific mitigation recommendations generated dynamically by the web logic.

Technology Stack

Built as a light, client-side Single-Page Application (SPA) using HTML5, Tailwind CSS (for responsive grid layouts and dark-mode rendering), Vanilla JavaScript (ES6+) for zero-latency calculation engines, Chart.js for interactive data visualization, and FontAwesome for semantic iconography.

5-Vector Granular Input Architecture:

1. *Energy & Utilities*: Electricity (with regional grid factors), Natural Gas, LPG Cylinders, Heating Oil, and Rooftop Solar mitigation logic (60% offset).
2. *Transport & Aviation*: Personal vehicle powertrains (Petrol, Diesel, Hybrid, EV), Public Transit, and Short/Long-haul Flight frequency tracking.
3. *Food & Diet*: 5 distinct dietary profiles (High Meat to Vegan), food waste discard multipliers, and local organic produce deductions (10%).
4. *Shopping & Goods*: Clothing spend, consumer electronics acquisitions, and baseline consumption profiles.
5. *Waste & Circular Economy*: Solid waste volume with deductions for paper/plastic recycling (−70 kg CO₂/yr) and home organic composting (−50 kg CO₂/yr).

Mathematical Calculation Model: Explicitly document the mathematical formulas driving the web software:

$$\text{Total Emission}(E_{\text{raw}}) = E_{\text{Energy}} + E_{\text{Transport}} + E_{\text{Food}} + E_{\text{Shopping}} + E_{\text{Waste}}$$

$$E_{\text{projected}} = E_{\text{raw}} \times (1 - \text{Pledge Target \%} / 100)$$

Key Interface Features:

- *Real-World Offset Equivalencies:* Converts raw tonnage into tangible metrics: Trees Required (Total kg/21.77), Gasoline Car Miles Equivalent (Total kg/0.404), and Smartphone Charges (Total kg/0.00822).
- *Dynamic Dark Mode & Persistence:* Native UI theme toggling and browser `localStorage` integration to prevent data loss on page reload.
- *Export Engine:* Integrated CSS `@media print` stylesheet allowing users to print or export an instant technical PDF summary.

Feasibility Analysis

- **Technical:** Highly feasible. The application relies on standard web development frameworks and open-source

emission factor databases (e.g., EPA or IPCC emission factors).

- **Economic:** Low development cost, utilizing open-source libraries (Tailwind, Chart.js) and static hosting (GitHub Pages / Vercel) results in **zero operational hosting costs**.
- **Operational:** The responsive web design ensures accessibility across desktop and mobile devices, lowering the barrier to entry for users.

System Architecture & Data Flow

- **UI/UX Layout:** Tabbed navigation structure separating user inputs into clear operational domains.
- **Data Processing Flow:** User Inputs → Event Listeners (`oninput`, `onchange`) → Core JavaScript Recalculation Loop → Canvas Redraw (`Chart.js.destroy()` & `new Chart()`) → DOM Node Updates (`innerText`).
- **State Management:** JSON serialization of input parameters saved to `localStorage` key `ecoPulseProState`.

Expected Environmental Benefits

- **Emission Reduction:** Identification of high-impact activities such as many short car trips helps individuals decrease their environmental impact by up to 10% - 20%.
- **Data-Driven Awareness:** It normalizes the discussion about carbon tracking, which results in collective data that would show sustainability trends at the community level.

Justification of the Proposed Solution

The interactive website is the best way to communicate this message, as the user receives instant feedback. Informational campaigns alone often don't help people break old habits, but a game-based calculator that is interactive helps keep users engaged while giving them a way to deal with the problem at hand.

Conclusion

The problem of high carbon footprints needs an easy-to-use, accessible solution. The Carbon Footprint Calculator can be seen as a viable and scalable IT solution for people to calculate and decrease their carbon footprint. It connects SDG 13 and SDG 12 in order to make a connection between international objectives and personal activities.

References

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- [3] Intergovernmental Panel on Climate Change, "2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories," *IPCC*, May 2019. [Online]. Available: <https://www.ipcc.ch/report/2019-refinement-to-the-2006-ipcc-guidelines-for-national-greenhouse-gas-inventories/>. [Accessed: Sep. 8, 2026].
- [4] U.S. Environmental Protection Agency, "GHG Emission Factors Hub," *US EPA Center for Corporate Climate Leadership*, Feb. 2024. [Online]. Available: <https://www.epa.gov/climateleadership/ghg-emission-factors-hub>. [Accessed: Sep. 8, 2026].

Appendix

EcoPulse Pro

Carbon Intelligence & Offset Analytics

SDG 12: Responsible Consumption

SDG 13: Climate Action

ISO 14001 & EPA FRAMEWORK ALIGNED

Quantify, Compare & Offset Your Environmental Footprint

Comprehensive calculation across home energy, transport, food, shopping, and waste vectors with real-time tree offset equivalencies.

EnergyTransportFoodShoppingWaste

REGION GRID FACTORMixed Grid (e.g., USA/Global Avg)

HOUSEHOLD SIZE1

Mobility & Flights

Weekly Vehicle Travel (km)60

Vehicle Powertrain TypeHybrid Vehicle

Public Transit (Bus/Train km/wk)0

Short Flights (< 3 hrs / year)10

Long Flights (> 3 hrs / year)0

REDUCTION GOAL SIMULATOR

15% Reduction Target

Simulate actionable lifestyle reductions to see immediate CO₂ and tree savings.

Reset DefaultsAuto-saved to browser storage

ANNUAL INDIVIDUAL EMISSION DENSITY

4.64 tCO₂e / year

✓ Moderate Impact (Global Avg)

Paris Target (2.0t)100%

REAL-WORLD OFFSET EQUIVALENCIES

213 Trees Needed to Offset / Yr

11,483 Gas Miles Driven

564.4 Phones Charged (k)

Category Emission Breakdown

EnergyTransportFoodShoppingWaste

Benchmark Comparison

ISO 14001 Targeted Action Plan

Dietary habits form your main carbon driver. Priority: Incorporate 2 plant-based days weekly and prioritize local produce to cut supply-chain transport emissions.

Export Technical Appendix Report

EnergyTransportFoodShoppingWaste

REGION GRID FACTORHydro/Nuclear (e.g., Norway/Iceland)

HOUSEHOLD SIZE4

Dietary Habits

Primary Diet ProfilePescatarian (Fish & Vegetarian - ~1.9 tCO₂e/yr)

Food Waste Discard LevelModerate (Occasional leftover waste - +15%)

Locally Sourced Organic Produce (~10% Diet Footprint)

REDUCTION GOAL SIMULATOR

15% Reduction Target

Simulate actionable lifestyle reductions to see immediate CO₂ and tree savings.

Reset DefaultsAuto-saved to browser storage

EnergyTransportFoodShoppingWaste

REGION GRID FACTORHydro/Nuclear (e.g., Norway/Iceland)

HOUSEHOLD SIZE4

Goods & Consumer Purchases

Clothing & Apparel (\$/month)e.g. 50

Electronics & Gadgets (\$/year)e.g. 500

General Consumption ProfileAverage Consumer (Standard purchasing habits)

REDUCTION GOAL SIMULATOR

15% Reduction Target

Simulate actionable lifestyle reductions to see immediate CO₂ and tree savings.

Reset DefaultsAuto-saved to browser storage

EnergyTransportFoodShoppingWaste

REGION GRID FACTORHydro/Nuclear (e.g., Norway/Iceland)

HOUSEHOLD SIZE1

Home Utilities & Fuels

Electricity (kWh/month)310

Natural Gas (\$ or Therms/month)0

LPG Gas (Cylinders 14.2kg/mo)2

Heating Oil (Liters/month)0

Rooftop Solar Installed (Offsets 60% Electricity Carbon)

REDUCTION GOAL SIMULATOR

15% Reduction Target

Simulate actionable lifestyle reductions to see immediate CO₂ and tree savings.

Reset DefaultsAuto-saved to browser storage

ANNUAL INDIVIDUAL EMISSION DENSITY

5.57 tCO₂e / year

▲ High Carbon Density

Paris Target (2.0t)100%

REAL-WORLD OFFSET EQUIVALENCIES

256 Trees Needed to Offset / Yr

13,785 Gas Miles Driven

677.5 Phones Charged (k)

Category Emission Breakdown

EnergyTransportFoodShoppingWaste

Benchmark Comparison

ISO 14001 Targeted Action Plan

Dietary habits form your main carbon driver. Priority: Incorporate 2 plant-based days weekly and prioritize local produce to cut supply-chain transport emissions.

Export Technical Appendix Report

EnergyTransportFoodShoppingWaste

REGION GRID FACTORHydro/Nuclear (e.g., Norway/Iceland)

HOUSEHOLD SIZE4

Waste & Circular Economy

Weekly Solid Waste OutputMedium Trash Bin (~600 kg/yr)

Actively Recycle Paper & Plastic (~70 kg CO₂/yr)

Home Organic Composting (~50 kg CO₂/yr)

REDUCTION GOAL SIMULATOR

15% Reduction Target

Simulate actionable lifestyle reductions to see immediate CO₂ and tree savings.

Reset DefaultsAuto-saved to browser storage